

Worksheet 2: September 4th

Problem 1. Prove that the sum of the first n positive odd integers is n^2 .

Problem 2. Define $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$.

(a) For a and b not both 0, rationalize

$$\frac{1}{a + b\sqrt{2}}$$

to show that this is an element of $\mathbb{Q}[\sqrt{2}]$.

(b) Prove, using the field axioms, that $\mathbb{Q}[\sqrt{2}]$ is a field.

Problem 3. Let $M(\mathbb{R}) = \{(a_1, a_2, a_3) \mid a_i \in \mathbb{R}\}$ for .

(a) Define $(a_1, a_2, a_3) + (b_1, b_2, b_3) = (a_1 + b_1, a_2 - b_2, b_3 - a_3)$ and $c * (a_1, a_2, a_3) = (ca_1, ca_2, ca_3)$. Do these operations make $M(\mathbb{R})$ a vector space?

(b) For $\mathbb{F}_2 = \mathbb{Z}_2 = \mathbb{Z}/2\mathbb{Z}$, define $M(\mathbb{F}_2) = \{(a_1, a_2, a_3) \mid a_i \in \mathbb{F}_2\}$. With the operations defined above, is this a vector space?

Problem 4. Let V be a vector space and A, B, C be subspaces of V such that $A, B \subset C$. Show that $A + B \subset C$ (so $A + B$ is the smallest subspace containing A and B).

Problem 5. Let $P = \{a_0 + a_1x + \dots + a_nx^n \mid n \text{ finite}, a_i \in \mathbb{R}\}$ be the space of real polynomials. Using the subspace criterion, show that the space $E \subset P$ of polynomials with only even powers of x and the space $O \subset P$ of polynomials with only odd powers of x are subspaces. Show further that $P = E + O$.

Problem 6. Let $U = \{(x, z - x, z) \mid x, y \in \mathbb{R}\} \subset \mathbb{R}^3$. Find V so $\mathbb{R}^3 = U \oplus V$.

Problem 7. Show that $\text{span}\{v_1, \dots, v_n\} = \text{span}\{v_1\} + \dots + \text{span}\{v_n\}$.